

Name.....

Roll No.....



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE FIRST INTERNAL TEST- MARCH 2023

Sixth Semester

CST362 PROGRAMMING IN PYTHON (Elective I)

Time: 2 Hours

Max. Marks: 50

PART A

Answer all questions (3 x 5 = 15 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1	Explain the input statement with an example. How the type conversion is done	3	K1	CO1&K3
2	Write a Python program to count number of even numbers and odd numbers in a given set of n numbers.	3	K3	CO1&K3
3	Write Python code for the following statements i) writes the text "PROGRAMMING IN PYTHON" to a file with name code.txt ii) then reads the text again and prints it to the screen.	3	K3	CO2&K3
4	Use list to read n names and print the names in alphabetical order	3	K3	CO3&K3
5	Let D = {'a':10,'b':20} be a dictionary. Write commands to a) Add a new key value pair('c':30) b) Update the value correspond to the key 'a' to 100 c) Remove the entry corresponds to the key 'b'	3	K3	CO3&K3

PART B

Answer any two questions (7 x 2 = 14 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
6	List and explain the different types of operators in Python	7	K1	CO1 & K3
7	Write a program to find the Area of a circle given its circumference	7	K3	CO1 & K3
8	Explain what goes on behind the scenes when your computer runs a Python program.	7	K2	CO1 & K3

PART C

Answer any two questions (7 x 2 = 14 Marks)

9	Write a python program to convert a decimal number to corresponding octal number	7	K3	CO2 & K3
10	Write a python program to remove all occurrence of a substring from a string	7	K3	CO2 & K3
11	Generate the Fibonacci series up to n (0 1 1 2 3 5....n)	7	K3	CO2 & K2

PART D

(Mandatory) (7 Marks)

12	Write a Python program to count how many times each character appears in a given string and store the count in a dictionary with key as the character.	7	K3	CO3 & K3
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Name.....

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**ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.***(An ISO 9001:2015 Certified College)***DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING****B. TECH DEGREE SECOND INTERNAL TEST- JUNE 2023**

Sixth Semester

CST362 PROGRAMMING IN PYTHON (Elective I)

Time: 2 Hours

Max. Marks: 50

PART A

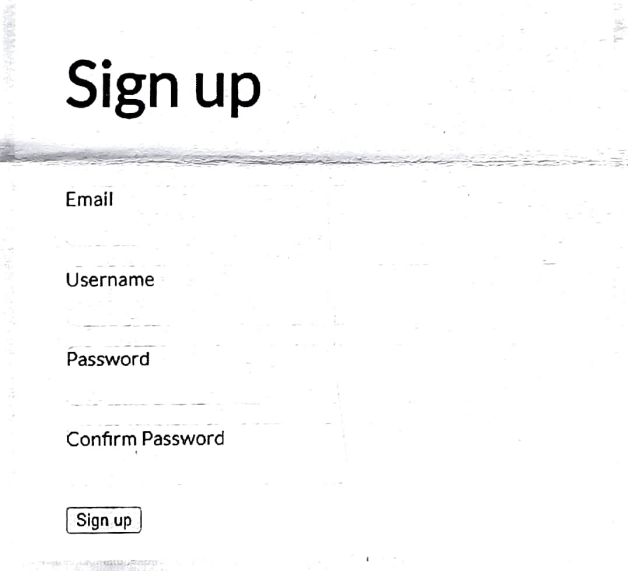
Answer all questions (3 x 5 = 15 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1	How to draw a flower shape using turtle in Python	3	K3	CO4 & K3
2	List any three image processing Python libraries.	3	K1	CO4 & K3
3	What is Hybrid Inheritance? How it is implemented in Python?	3	K2	CO5 & K3
4	Explain the terms accessors and mutators with regard to Python class definition.	3	K2	CO5 & K3
5	What is Flask in Python? What are its basic components?	3	K1	CO6 & K3

PART B

Answer five questions (7 x 5 = 35 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
Answer any two questions				
6	What are exceptions? How does Python catch it? Illustrate the usage.	7	K2	CO5 & K3
7	Write a Python program to create a class representing a stack data structure. Include methods for pushing, popping and displaying elements.	7	K3	CO5 & K3
8	Write a Python program to create an abstract class Animal with abstract methods eat() and sleep(). Create subclasses Lion, Tiger, and Deer that extend the Animal class and implement the eat() and sleep() methods differently based on their specific behavior.	7	K3	CO5 & K3
Answer any two questions				
9	Write a NumPy program to get the number of non-zero elements in an array. Expected Output: Original array: [[0 10 20] [20 30 40]] Number of non zero elements in the above array: 5	7	K3	CO6 & K3

10 ✓	Temperature(°C) on different dates is stored in a CSV file as 'Weather_data.csv' with the fields date, temperature and humidity. 1. Draw a plot of the weather report with date as the x-axis and temperature as the y-axis. 2. Draw a scatter plot of the weather report with date as the x-axis and humidity as the y-axis. Give appropriate titles and labels in the plot.	7	K3	CO6 & K3																				
11 ✓	Write Python program to write the data given below to a CSV file <table> <tr> <th>Reg No</th><th>Name</th><th>Physics</th><th>Chemistry</th><th>Maths</th></tr> <tr> <td>SJC24CS067</td><td>Rahul</td><td>78</td><td>67</td><td>95</td></tr> <tr> <td>SJC24CS069</td><td>Hari</td><td>90</td><td>88</td><td>77</td></tr> <tr> <td>SJC24CS089</td><td>Christy</td><td>66</td><td>88</td><td>55</td></tr> </table>	Reg No	Name	Physics	Chemistry	Maths	SJC24CS067	Rahul	78	67	95	SJC24CS069	Hari	90	88	77	SJC24CS089	Christy	66	88	55	7	K3	CO6 & K3
Reg No	Name	Physics	Chemistry	Maths																				
SJC24CS067	Rahul	78	67	95																				
SJC24CS069	Hari	90	88	77																				
SJC24CS089	Christy	66	88	55																				
Mandatory Question																								
	Design a simple signup form using GUI as shown in the figure 																							
12 ✓	Email Username Password Confirm Password Sign up	7	K3	CO4 & K3																				

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R, S) / S4 (PT) (R, S) Examination June 2023 (2019 Scheme)

Course Code: CST362**Course Name: PROGRAMMING IN PYTHON**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- 1/ Jack says that he will not bother with analysis and design but proceed directly to coding his programs. Why is that not a good idea? (3)
- 2/ Write the output of the following python statements : (3)
i) round(12.57) ii) 5//2 iii) int(6.5)
- 3/ Write the output of following python code : (3)
S = "Computer"
print(S[:2])
print(S[::-1])
print(S[:])
- 4/ Write a recursive function in python to find GCD of two numbers. (3)
- 5/ Illustrate the function of following methods in turtle (3)
i) turtle.setheading(0) ii) turtle.forward(50) iii) turtle.left(90)
- 6/ Describe two fundamental differences between terminal-based user interfaces and GUIs. (3)
- 7 Explain what the `__str__` method does and why it is a useful method to include in a class. (3)
- 8/ Compare multiple and multilevel inheritance (3)
- 9/ Write the output of the following python code: (3)
import numpy as np
arr1 = np.arange(6).reshape((3, 2))
arr2 = np.arange(6).reshape((3, 2))
arr3 = arr1 + arr2[0].reshape((1, 2))
print(arr3)

- 10 What is the difference between **loc** and **iloc** in pandas DataFrame. Give a suitable example. (3)

PART B

Answer one full question from each module, each carries 14 marks.

Module I

- 11 a) Write a python program to find the sum of the cosine series $1 - x^2/2! + x^4/4! - \dots$ (7)
- b) Write a python program to find X^Y or **pow(X,Y)** without using standard function (7)

OR

- 12 a) Write a python program to generate the following type of pattern for the given N rows where $N \leq 26$. (7)

A	1	1				
A B	2	1	2			
A B C D	4	1	2	3	4	
A B C D E	5	1	2	3	4	5

- b) Write a python program to generate prime numbers within a certain range (7)

Module II

- 13 a) Assume that the variable `data` refers to the string "Python rules!". Use a string method to perform the following tasks: (6)
- Obtain a list of the words in the string.
 - Convert the string to uppercase.
 - Locate the position of the string "rules".
 - Replace the exclamation point with a question mark.
- b) Write a code segment that opens a file for input and prints the number of four-letter words in the file (8)

OR

- 14 a) Assume that there is a text file named "numbers.txt". Write a python program to find the median of list of numbers in the file without using standard function for median. (10)
- b) Use higher order python function **filter** to extract a list of positive numbers from a given list of numbers. You should use a lambda to create the auxiliary function. (4)

Module III

- 15 a) Write a python function to draw a square using turtle graphics (5)
 b) Write a python function to convert an image to black and white using image processing methods. (9)

OR

- 16 a) Write a python function to draw a hexagon using turtle graphics (5)
 b) Write a python function to shrink an image by a given factor. The function should build and return a new image which is a smaller copy of the argument image, by the factor argument. (9)

Module IV

- 17 a) Write a Python program to create a class called as **Complex** to model complex numbers and implement `__add__()` and `__mul__()` methods to add and multiply two complex numbers. Display the result by overloading the `+` and `*` operator. (9)
 b) Explain multiple inheritance in Python with a suitable example (5)

OR

- 18 a) Write a Python program to create a class called as **Rational** to model rational numbers and associated operations. Implement the following methods in the class. Use operator overloading. (9)
 1. `Reduce()` – to return the simplified fraction form
 2. `__add__()` - to add two rational numbers
 3. `__lt__()` - to compare two rational numbers (less than operation)
 b) What is Exception handling? Write a program that opens a file and writes "Hello Good morning" to it. Handle exceptions that can be generated during I/O operations (5)

Module V

- 19 a) Write a code segment that prints the names of all of the items in the current working directory. (5)
 b) Write a python program to create two numpy arrays of random integers between 0 and 20 of shape (3, 3) and perform matrix addition, multiplication and transpose of the product matrix. (9)

OR

- 20 a) Write a Python program to write the data given below to a CSV file named **student.csv** (4)
`fields = ['Name', 'Branch', 'Year', 'CGPA']`

```
rows = [ ['Nikhil', 'CSE', '2', '8.0'],  
          ['Sanchit', 'CSE', '2', '9.1'],  
          ['Aditya', 'IT', '2', '9.3'],  
          ['Sagar', 'IT', '1', '9.5']]
```

- b) Consider the above **student.csv** file with fields Name, Branch, Year, CGPA . (10)
Write python code using pandas to
- 1) To find the average CGPA of the students
 - 2) To display the details of all students having CGPA > 9
 - 3) To display the details of all CSE students with CGPA > 9
 - 4) To display the details of student with maximum CGPA
 - 5) To display average CGPA of each branch
